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Development and Implementation of Research Conference Portal with Archiving for PHINMA- Saint Jude College Manila

A Capstone Project
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College of Information Technology Education
Manila

In Partial Fulfillment of the
Requirements for the Degree of
Bachelor of Science in Information Technology

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ABSTRACT

This paper presents a web-based Research Conference Portal with archiving abstract paper, payment and conference live streaming. To facilitate the preparation until the end of the conference day. It is used to support the multiple conference organizers in the management participants' paper, reviewing the process of the entire conference. With the conference live streaming consumers are certain that you are hosting a great event, which provides them reason enough to invest the time, money, or necessary resources to join you on location the next time they receive an invitation. In order to present the research study of the author by submitting a work paper until it is completed, the release paper has been approved by the admin and it will be released to the public after the payment transaction. In exchanging the ideas between the participant and the current researchers.

Chapter I. The Problem and Its Background

1.0 Introduction

Conference is a meeting for researchers to present, discuss, share, identify important problems, details and formulate for future research direction on specific ideas. It provides a true learning environment. However, organizing a conference entail plenty of time-consuming work. For instance, collecting paper submissions from authors, assigning paper submitted to the reviewers, notifying authors about whether the paper submitted is rejected or accepted. This process will surely consume a lot of precious time and complicated work if a conference is conducted manually. Unfortunately, there are no reports from literature that are seriously looking into how to produce an efficient system in managing the entire process of conference (Sultan et al., 2009). The conference event has grown constantly each year and the complexity of handling this year's amount of predicted data overwhelmed the human resources available, and also raised serious issues in the revision and submission processes (Brad, 2008)

A conference is divided into several phases. The first phase is call for papers, which includes getting the targeted emails such as previous participants of the current conference in addition to many other researchers and institutions. The second phase is submitting articles from the participants to the conference managerial board. Then, these articles are classified and a pre-decision of denying or accepting them is taken. The revision, which is the third phase, comes after receiving the article. Bidding and automatic or manual paper assignments are processed in this phase. Paper assignment to reviewers can be done automatically or manually. The decision is taken on accepting or rejecting the article and the authors are informed after the notification of acceptance date as announced in the conference web-pages or brochures. Authors, whose papers have been accepted, are informed to send their camera-ready (revised and well formatted version of their articles) back to the conference system. Thus, the fourth and final phase is collecting the camera-ready articles and producing the conference program. Then, the complete set of camera-ready articles is published. The previous mentioned phases can be managed manually, by email, or in an automated way. Traditional or manual conference management systems are hardly seen. Current conferences are either managed by email or in an automated way Awad et al. (2015).

Research Conference Portal with Archiving for CITE Department in PHINMA-Saint Jude College Manila is a web-based that will be used as a tracker and to facilitate the workload of the paperwork submission details for conference. An online website portal and mobile app that will support the operation of conference information. The CITE department administrator of PHINMA SJC Manila will check and monitor the paperwork submission for approval. The system provides payment transaction details throughout the process of each paperwork submission. The Conference Portal system has a live streaming on the same day of the conference. The participant can watch the conference online using the system. In addition, it will reduce the workload and manpower of the conference organizer, faculty and staff in the school. The easy and simplified way of managing conferences will help the institution to hold paperwork efficiently and proceed according to the agreed plan of faculty members with the consent of the conference organizer.

1.1 Project Context

1.1.1 Purpose and Description

An efficient tool for handling conference and paper submission through an internet solution aids a business in achieving better results and effective management for this goal. The goal of this project was to create a web-based solution that would make it simple to manage the conference process and reduce the amount of work required from the conference organizer, attendees, and user. Xampp software, HTML, CSS, PHP, bootstrap, and other technologies were used to develop the user interface. Furthermore, the Bootstrap framework was used to make the website user-friendly.

The number of users who access and register on an online conference portal solution is strongly correlated with the number of users. The solution's main goal is to use the internet to automate and manage various occurrences.

1.2 Objective of the Study

1.2.1 General Objective of the Study

To develop a conference portal with archiving for PHINMA -Saint Jude College to store, and provide a centralized platform to manage all conference tasks and details.

1.2.2 Specific Objectives of the Study

Analyze data to improve the future and better cater to the needs and preferences of attendees. The system's specific objectives are listed below:

1.2.3 To organize, store, and make accessible all digital content related to the conference, such as presentations, discussions, and conclusions for posterity and for those unable to attend and archive paperwork of the author and researcher.

1.2.4 Effectively handle the financial transactions related to the conference, from attendee registration fees to vendor payments.

1.2.5 The system will focus on providing an intuitive and seamless experience for users, participants, speakers, and stakeholders.at all stages of the process, from conference creation to record event details.

1.2.6 Users can remark and engage in real time by capturing and streaming live audio and video.

1.3 Scope and Delimitation of the Study

1.3.1 Scope

The scope of the proposed Development and Implementation of Research Conference Portal with Archiving for PHINMA -SJC Manila system includes the processing of user profile, payment transaction, abstract paper and live streaming involving the participants of PHINMA-SJC. The administrator has an access to review and approve or reject the submitted paper in the conference portal with archiving. The administrator of CITE Department of PHINMA- Saint Jude College Manila, they must sign-up and log-in, on the website. Each of the users need to type their username and password. These two enable them to access their accounts. Conference Portal that can make event planning simple, for the preparation and organize a conference, on site or virtual. The submission, reviewing of contribution, scheduling of the conference program, registration, administration, invoicing of participants, managing of communication between authors and participants. The conference portal is more flexible and suitable for the event of the CITE department in PHINMA- SJC. Conference provides organizers a simpler way to manage events. It is beneficial

for gathering submissions, scheduling presentations and speakers, and managing talks and speakers. Matching the interests of the reviewers with the article themes.

The following will be covered for the proposed Research Conference Portal with Archiving:

1. Role-based access and User authentication

Admin- Can edit the conference creation, verify the new user, approval submitted abstract paper and facilitate the entire conference session.

Participants- The person who attended in the conference, view conference session

Author- The person who submitted the abstract paper and view the upcoming and ongoing conference event.

Reviewer- Reviewing the submitted abstract paper by the author

2. Conference Setup and Management

Attendees and presenters- this translates to a well-organized, smooth-running event.

Organizers- top-down view of the conference landscape, providing the agility to make real-time decisions and adjustments. In essence, the Conference Setup and Management feature ensures that the blueprint of the conference is flexible, catering to the dynamic nature of such events.

3. Submission Management

Serves as a critical bridge between conference presenters and organizers. At its core, it facilitates the submission of research papers, abstracts, or posters by authors and presenters. Once these documents are submitted, they enter a review pipeline, often designed to be blind, to ensure unbiased evaluations.

4. Archiving and Repository

This section ensures that all past conference materials, from presentations to research papers and abstracts, are systematically stored, categorized, and made easily accessible for future reference. A well-structured archiving system not only preserves these invaluable resources but also equips users with a searchable database, allowing them to tap into past wisdom, draw parallels, or build upon prior knowledge.

5. Live Streaming and Virtual Participation

The portal allows attendees to partake in real-time discussions, attend sessions, and even engage in side conversations, much like they would in a physical setting. Virtual "rooms" or "halls" are designed to host different sessions, ensuring participants can hop between topics of interest seamlessly. Additionally, live chat and Q&A capabilities ensure that the dialogue between presenters and attendees.

6. Payment and Billing System

From attendee registrations to vendor space bookings, the system provides instant, transparent, and hassle-free payment options. By integrating with trusted payment gateways, it ensures that all transactions are not only swift but also safeguarded against potential security threats.

7. Notification and Communication

Ensuring that all stakeholders—be it attendees, presenters, reviewers, or organizers—are consistently informed and engaged. In an environment where timely information is paramount, this feature guarantees that no crucial detail goes unnoticed. Whether it's about paper acceptance, session updates, venue changes, or imminent deadlines, the system dispatches real-time alerts via email or in-app notifications, ensuring everyone is always on the same page.

8. Mobile App Integration

The app, optimized for smaller screens, offers a user-friendly interface, allowing users to easily navigate through sessions, engage in discussions, or even network with peers, all while on the move.

9. Security and Data Protection

This feature ensures that every piece of data, from personal attendee details to sensitive research submissions, is safeguarded with state-of-the-art encryption and security protocols.

1.3.2 Delimitation

The study is limited to signing up and log in, the admin are the only users allowed to access the actual system in which they can edit, add and delete conference, approve or reject the paper work and add admin access. While the researcher and author can view the upcoming conference and are allowed

to submit the paperwork through the portal system. Only authorized personnel can access the system. This study is not applicable offline and does not display the announcement and information about the conference portal if it has no internet connection.

The proposed Development and Implementation of Research Conference Portal with Archiving for PHINMA-Saint Jude College Manila limits its coverage to the following:

Type of Users

The portal may focus on researchers, academics, and professionals, excluding students or the general public.

Geographical Restrictions

The platform might be developed with a specific region or language in mind, limiting its usability on a global scale.

Field of Study

The portal may cater specifically to certain disciplines, such as CITE fields, and may not be suitable for conferences in other field such as CAHS, CAS, CMA, COED & etc.

Size of the Conference

The portal may be optimized for small conferences and might not be suitable for medium to large-scale or workshop.

Platform Integration

The portal might only support certain third-party tools for live streaming or payment processing, thereby excluding others.

Device Compatibility

The portal might be designed primarily for desktop use, with limited functionalities on mobile devices and have an internet connection.

Duration of Archiving

The archiving feature may have a set duration, post which older papers or presentations might be deleted or archived offline.

Modes of Communication

The portal may focus on digital communication (emails, in-app notifications) and may not support SMS or physical mail.

Single Event Focus

The platform might be designed to manage one conference at a time, making it less suitable for organizations that host multiple simultaneous events.

Feedback Mechanisms

The portal might have predefined feedback mechanisms, and might not support open-ended or custom feedback forms.

1.4 Significance of the Study

This study aims to help and benefit to the following people in various ways, which will be discussed below:

Administrator

They will have an easier time organizing and managing the task list, the timeline of a conference, the flow of the program, inclusion details, user management, and all the information for the management of the conference. Administrators can also benefit in arranging the upcoming conference and allocating them for a specific day of paper submission. It can lessen misunderstandings and confusion caused by multiple paperwork taking place in the same location on the same day.

Conference Organizer

It helps them to deliver high quality events with less stress and greater efficiency and success. The system helps organizers save time, reduce manual efforts, enhance participants satisfaction, and ultimately create more successful and memorable conferences.

Users

The system offers several benefits to users, including streamlined registration processes, easy view of the conference information, personalized experiences, and efficient communication. It enhances the overall conference experience by simplifying logistics, reducing manual tasks, and enabling participants to engage more effectively with the conference. The system archives the author's work in a centralized database which could be accessed by authorized users.

School

The system helps the school to provide a conference management portal system that can simplify the registration, planning and submission of paperwork. Ensures all the campus conferences are easily scheduled and without miscommunication. It tracks logistic management, floor plan, venue sourcing and data analytics which increase the productivity of educational institutions. School authorities can save their valuable time by keeping the current research of a student and facilitating a

variety of conferences in a single location. The administrator of the department may proceed with their plan and carry it out with the portal system.

Proponents

In this stage, researchers can have a solid foundation on which they can build their ideas and plans. Enhance their knowledge and skills in system analysis that they have chosen.

Future Researcher

The ideas offered here could be used as reference data for new researchers or to test the validity of other similar results. This study will serve as a guide for their research study.

CHAPTER II: REVIEW OF RELATED LITERATURE AND STUDIES

This chapter indicates the ideas and concepts relevant to the present subject relating to the other studies and is discussed to provide the foundation of the proposed study. In order to find the correct method and procedures, careful review of literature and studies must be done for the development of the study.

2.1 Theoretical Background

Technology Acceptance Model

The technology acceptance model (TAM) is an information system theory that explains how to encourage users to accept and utilize new technology. Technology Acceptance Model or TAM, perceived usefulness and perceived user-friendliness are the two elements that affect whether potential users will embrace a computer system. This model's emphasis on the perceptions of the potential user is its defining characteristic. That is to say, even while a technology product's inventor may think it is practical and user-friendly, the product won't be accepted by potential customers until those consumers also have the same opinions (Davis, 1989). Although intentions often result in behavior, the Technology Acceptance Model asserts that this is not always the case. For instance, a person could have good intentions but not actually use internet treatment. The degree of the correlation between intentions and conduct is influenced by a number of variables (Georgia et al., 2021).

The proponents choose the technology acceptance model to know and understand the amount of users' adaptation in using the system based on their attitudes. They would like to know how the user behaves and adapts to the system and also how the system is useful for them.

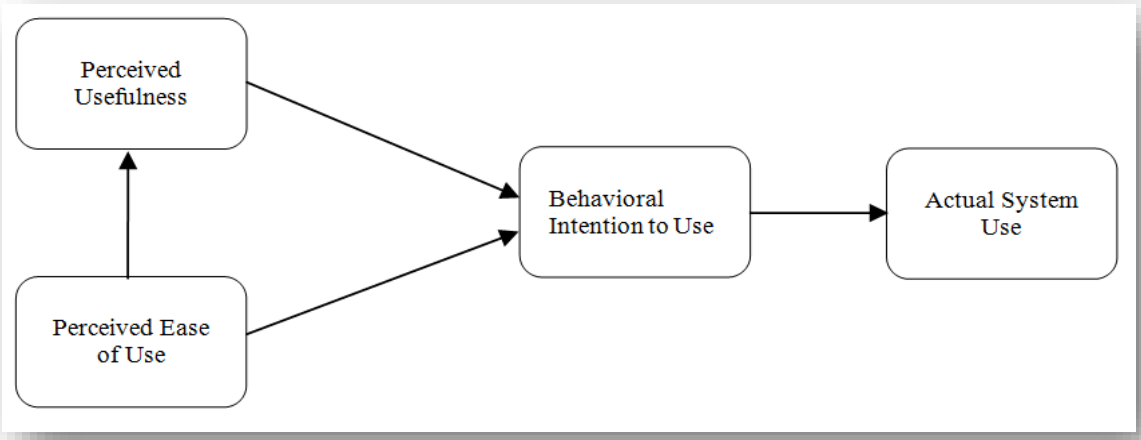

Figure 1: Technology Acceptance Model Diagram

2.2 Review of Related Literatures and Studies

According to Hankskin & Nel (2013), it was vital to discover why the current website was created and what its flaws were in order to decide how to effectively construct an updated conference portal. In order to further inform the construction of the new student online portal, a number of previous conference portals were evaluated in terms of functionality and GUI design. The goal of this study was to design a checklist to help with the creation of an online conference management portal. In order to do this, the current portal was examined, and data on the prerequisites for a productive conference portal was obtained. A new conference portal was created using the information provided.

According to Zaibon et al., (2010), a portal was created in the beginning to aid users in their search for information on the World Wide Web (WWW). While a conference is a type of platform for talking about a big topic. The goal of the project is to create a web-based information hub that focuses on promoting new conferences. A conference portal is therefore thought to improve communication between companies and users. The traits and capabilities of a portal that were gathered and discovered from various research sources are discussed in this project. Potential users constructed and further tested a prototype system. In essence, the earliest online portals were just simple search engines that provided a list of websites including phrases and/or keywords entered by the user. Following the development of web-based search users required more from search engines than a list of thousands of websites that had the selected search terms. The creation of web portals is the logical next step in this. Then, the features and functionalities were combined to create a conference portal prototype that would be useful to both prospective participants and organizers. In essence, this study involves five phases, including creating a conceptual framework, creating a system architecture, conducting system analysis and design, and constructing a prototype system, and then monitoring it and assessing it. Studying the literature and contrasting various lists of features and functionalities were identified for the conference portal. The collection of capabilities and features can be leveraged to create additional portals for conferences.

A conference's details should be disseminated internationally to reach a bigger audience (Mohammed, 2022). For many individuals, physical interaction is insufficient and won't assist in

dispelling people's skepticism. We suggest managing registrations online framework for academic conference management in any institution or university to address all of the conference's use of the physical system's restrictions. For the registration of academic conferences, this application is designed and developed. There is an automatic system necessary to manage all the data on a physical basis. An application with similar functionalities is called "easy chair". In comparison to our application, however, the easy chair application provides a greater range of functions. Only the registration system is targeted in our application. Therefore, this application is not a competition for any of the other similarly submitted applications. The information can be viewed by the public, interested parties can sign up for this application and communicate with the conference participants, receive suggestions, or advice from experts in a given field, etc. Users must be signed in to make it happen with the application for which he must log on to perform a series of operations in that application. Users will have the option to edit their personal data, and if they want to find experts in a field of interest, they can do so. They may use this application if necessary for communication with them. In order to create this application, .Net is being used by proponents for creating a GUI and performing various functions of the application. They have coded with HTML, CSS, BOOTSTRAP, JAVAScript and jQuery in SQL Server Administration Studio to store data and ASP.Net 4.5 Framework as the framework.

In all aspects of our lives, web development has had an impact on communication, access to information, job seeking, community activities and so forth. To individuals who are connected to the Internet, a web portal is an initial point of entry on the World Wide Web. Web portal as a type A knowledge management system offers a rich environment in which information can be shared, searched and communicated via email or the provision of content to users. The study will focus on Web portals, which play an essential role in school knowledge management systems. In general, the web portal is a gateway to accessing information and searches, as well as an audiovisual and communication platform for target users. In order to help students access the right sources and information, it will be important for schools to provide a focus on knowledge sharing. Web portals are available in different forms, with various features that bring advantages to clients. Every customer's personal distinctive meaning of the Web Portal may be apparent to him or her. Essentially, a portal is an access to the Internet network that has been made possible by available resources, via an intranet, extranet or internet. This definition may be changed by the Basic Web page, as could an uncertain site that included a large size. The number of Web pages on this site. The early portals were very basic, giving individuals a static perspective on the substance's contents from several points of view. The

web portal is, for the most part, a tool that allows clients to access information from alternate sources in an integrated manner. Web portals offer a variety of services, for example, email, news, stock costs, information, and diversion, depending on the way in which the portals host establishments operate, in addition to the standard Internet searcher.

However, some courses, such as those requiring a significant research component, present significant obstacles to the delivery of online courses, while the content of most traditional courses can be delivered online and learning outcomes can be achieved through the adoption of equivalent face to face pedagogic approaches. Australia's biggest university has set up and launched a research portal in order to overcome these limitations. This paper sets out the context, design, deployment and possible transferability of an Integrated Online Research Environment that enables its users to perform all aspects of research on the Internet. The development team has identified the parts of the user's needs that would have to be met, made a technical specification analysis and examined existing solutions. The Monash University Psychology Research Portal was designed, tested, developed, tested for user experience, implemented, and reviewed. In order to provide a user with access to the supporting information and integrated, the Research Portal shall be structured according to normal research sequences. capacity for supervision of studies, acquisition of participants' information and data collection, analysis and storage. The lessons learnt in developing the Research Portal include recognition of the necessity to use a broad variety of expertise at different stages of development, identification of various potential users and encouraging them to have an active role in its development and evolution. pragmatism and flexibility in the creation and dissemination of a research portal.

The thesis stage is one of the most important periods in a college education. A student needs to do that last major project before he or she can graduate. In this regard, it is important to ensure that procedures are correctly and effectively carried out at this stage of the process so as to allow students to present an excellent thesis. This paper presents the development of a Portal with Electronic Document Management system for the Department of Information Technology at The University of Manila College of Computer Studies. This system shall apply to the entire thesis process, as well as document management for a variety of thesis documents. It is the provision of a portal that can help to better track and carry out the thesis cycle that is at the heart of this system. Time has emerged as the most important problem of all that had been collected in developing this system. In order to gather the relevant data and comments in order to find a solution, the Proposers conducted

interviews with several persons involved. Following this information, the Proponents have come up with an alternative solution whereby students and faculty will be able to discuss their ideas through a single channel. The proponents were able to maximize the learning time of students and teachers in their work on this thesis by using such a system. Other tasks that may contribute to the productivity of faculty members and students can be made use of these time savings. A test for users has been carried out at the end of the development phase. User acceptance tests carried out have shown positive results. The problems identified in the paper have been dealt with by the system. A positive result of the user acceptance test was obtained. The issues set out in the report were dealt with by the system. Consideration has also been given to improving the system based on user feedback.

2.3 Technical Background

Technicality of the project

The proponents have used web-based program wherein the users can access it through any digital platforms and mobile application as long as with internet connection. The Conference Portal with Archiving for PHINMA-SJC is about the presentation and discussion of the paperwork of the registered participants in the conference for easy and convenient access by the admin of the department faculty and participants without using the traditional process. These are some of the technical terms that are being used in our project: XAMPP, PHP, HTML, CSS, Database MySQL, Functionality – Web application, PhpMyAdmin. Some of the terminologies being stated above are also the technology being used in our project.

Details of the technologies to be used

This project will be available in any technology such as mobile phones, laptops, and computers to run the system. In developing the project, the following technology tools will be used:

PHP – is a general-purpose scripting language that is especially suited to server-side web development where PHP generally runs on a web server. It can also be used for command-line scripting and client-side GUI applications.

Xampp – Developers will use this as a development tool, to allow website designers and programmers to test their work on their own computers without any access to the Internet.

MySQL – is an open-source relational database management system. MySQL will be used by the developer, because it is one of the system development compatible databases.

HTML – This is the language that Web pages are written in. As far as computer languages go this is the easiest to learn. You can create a Web page without it using a Web page editing program but the program will still use HTML to create the page.

CSS – Cascading style sheets are used to format the layout of Web pages. They can be used to define text styles, table sizes, and other aspects of Web pages that previously could only be defined in a page's HTML. CSS helps Web developers create a uniform look across several pages of a Web site

2.3.1 Conceptual Framework

Presented below is the conceptual model of the proposed system for PHINMA-Saint Jude College Manila.

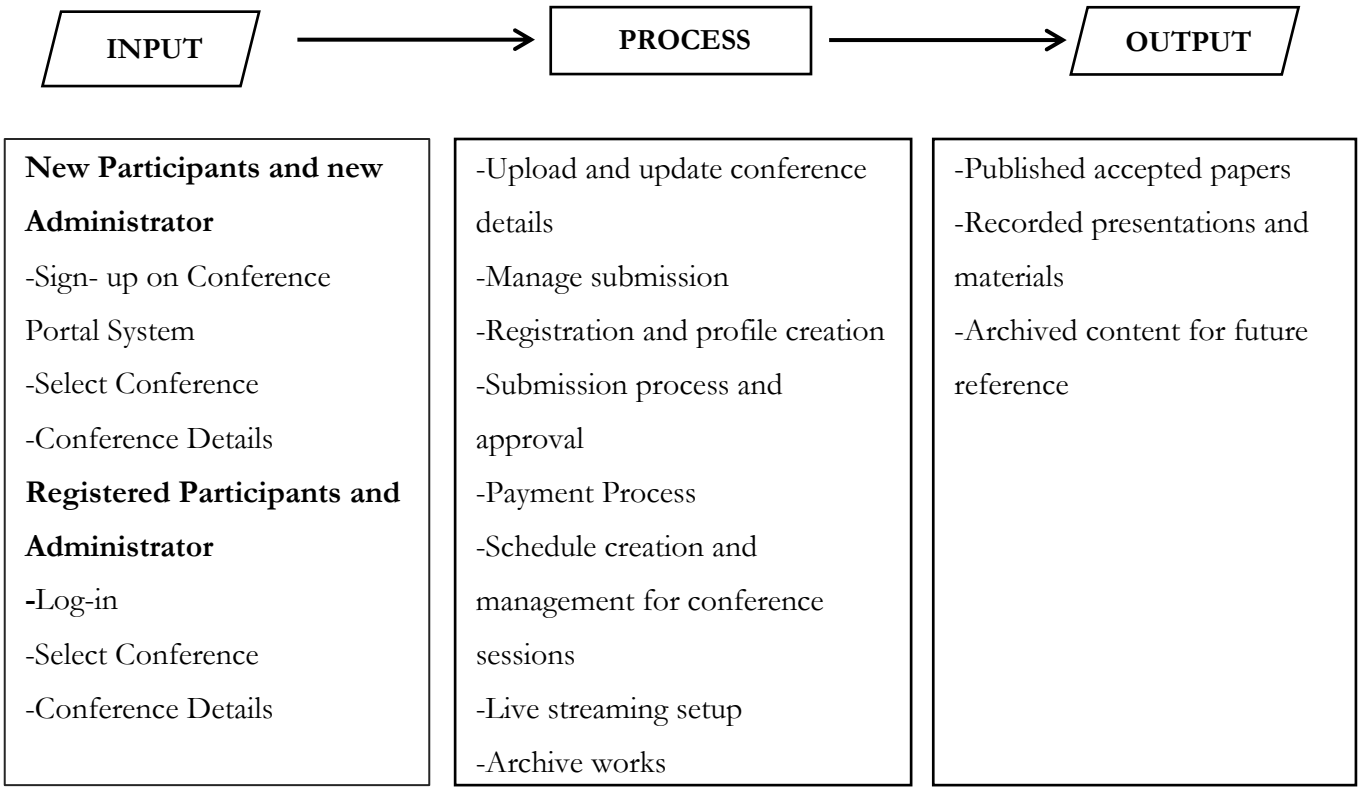

Figure 2: Process flow of the Proposed System

Definition of terms

The following words are defined theoretically and operationally to give a common frame of reference and to assure understanding of the contents of this study.

Abstract - An abstract is a brief overview of a research paper or a complete thesis. It is an original composition, not a paraphrase. An abstract must be self-contained and make sense without further reference to outside sources or the article itself.

Conference - is a meeting of two or more persons for discussing matters of common concern about the research paper. The admin is in conference with the participants. Usually formal interchange of views and consultation.

Live streaming - when the streamed video is sent over the Internet in real time, without first being recorded and stored.

Participants - someone who takes part in an event or activity.

Paperwork - is the routine part of a job which involves writing or dealing with letters, reports, and records.

Payment Transaction - an act, initiated by the Payer or on his behalf or by the Payee, of placing, transferring or withdrawing funds, irrespective of any underlying obligations between the Payer and the Payee. Can be made in the form of services exchanged, cash, check, wire transfer, credit card, debit card, or cryptocurrencies. The series of actions that occur when digital payment transactions are initiated by a business.

Portal - a web-based platform that compiles data from various sources into a single user interface and shows consumers the information that is most pertinent to their context.

System - any framework of software that facilitates the collection, storage, organization, and distribution of information. A computer system is a programmed electronic device that can accept input, store data, retrieve, process, and output information.

Web-based - an external application that is accessed via a web browser over the Internet. A Web-based application refers to any program that is accessed over a network connection using HTTP, rather than existing within a device's memory. Web-based applications often run inside a Web browser.

Workload - The amount of work or of working time expected or assigned. Students with a heavy workload. The amount of work performed or capable of being performed usually within a specific period.

Chapter III. Research Design and Methodology

This chapter shows research design used to serve as the guiding blueprints for the entire study. The research design is the strategy that defines the connection of the research question and its method, setting the foundation for how the study conducted. Ensuring the investigation remain coherent, systematic and relevant to its objectives. Ensures the study's direction, purpose is clear, credibility and validity of findings.

3.1 Research Design

In this study shows quantitative survey research design using a questionnaire. Survey research method used to describe and explain the respondents experience and share their opinion about the research topic that provide a detailed knowledge. Allowing the researcher to gather information and analyze the data for the various objectives. This quantitative research design allows researchers to provide respondents with anonymity and confidentiality, encouraging honest responses. Quantitative survey research design is a systematic approach focused on gathering numerical data from a predefined group to identify patterns, relationships, and trends within that group. Using structured tools such as questionnaires or interviews, this design ensures uniformity in data collection, allowing for clear comparisons and statistical analysis. In order to determine the outcomes of the participants, researcher, and the conference chair perception about challenges of the current conference system of PHINMA- Saint Jude Manila. This study will serve as understanding of the conference portal's performance and areas of improvement.

3.1.1 Software Architecture

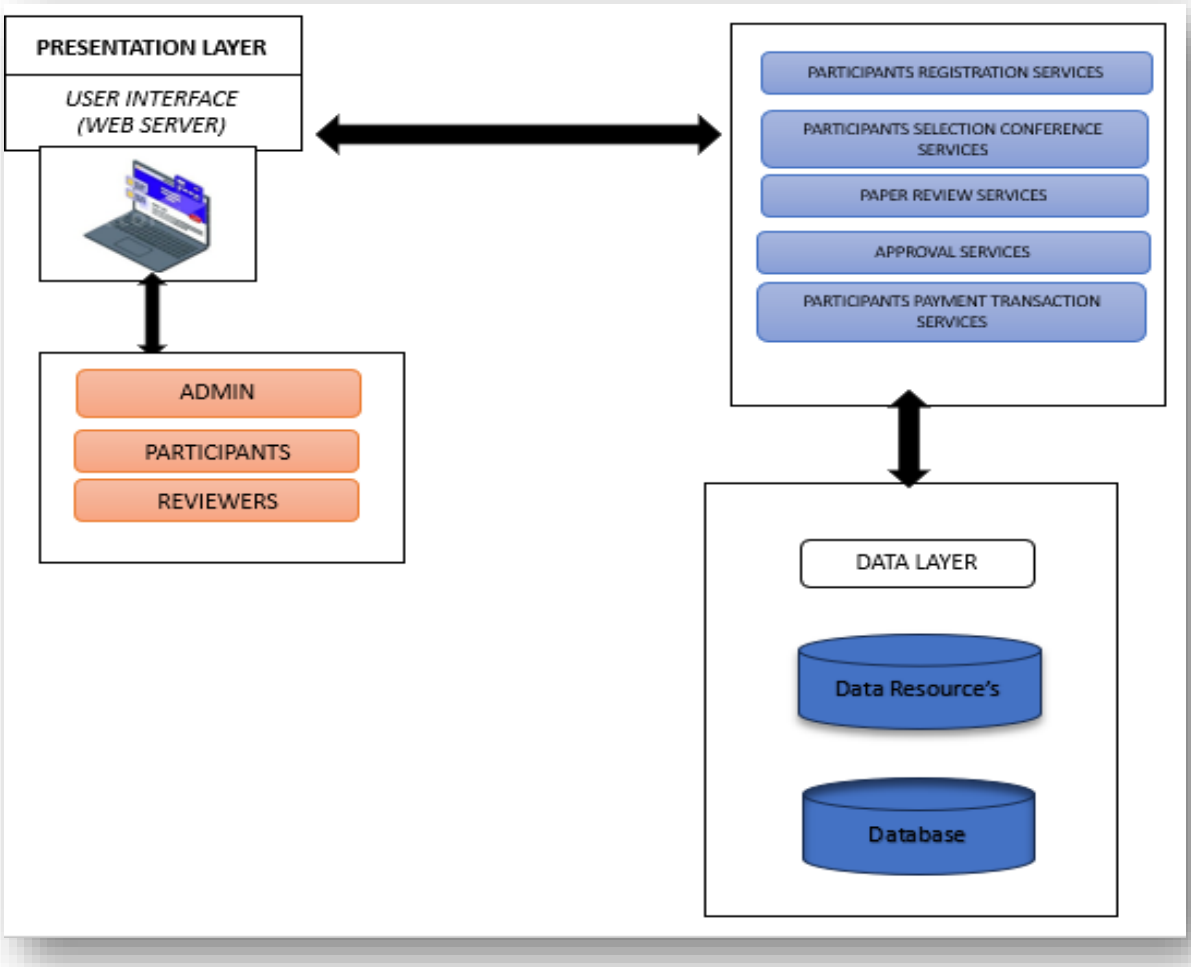

Figure 3: Architecture of Conference Portal

In figure 3. Present the user interface, a web server with the 3 users. The Conference Portal supports the participants, registration, submission, review, payment phases of the conference process. The Conference Portal System architecture is decomposed into three actors: The presentation layer, process from registration to payment process and data layer. Each of these actors is linked to the system through an is-part-of relationship are informed that their papers have been received and have been assigned a submission number.

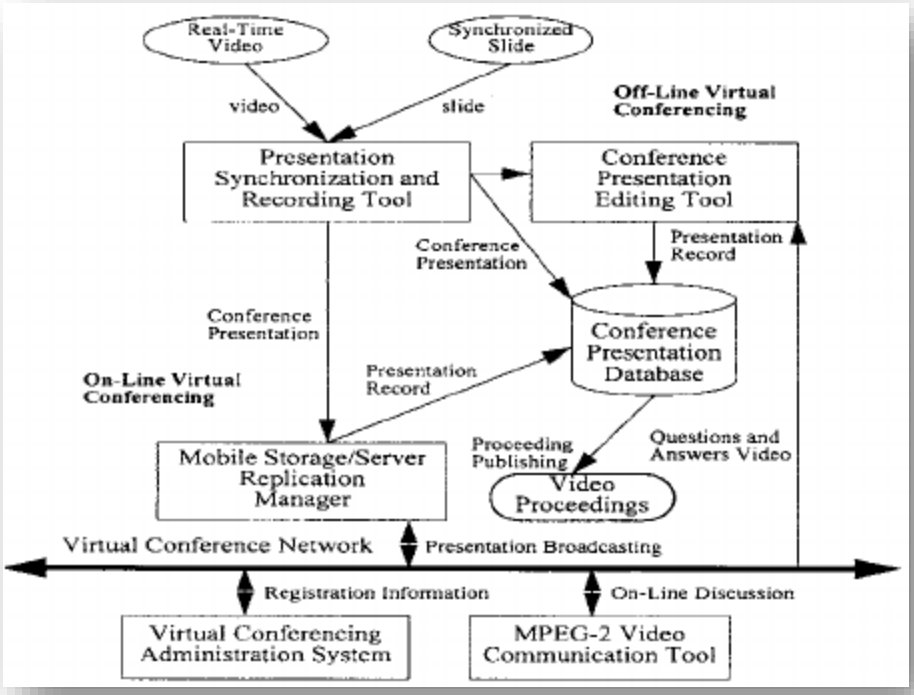

Figure 4: The software components of sample virtual conferencing system. (source: Wang (et.al) 2001)

The initial version of the Conference Management System supports the submission, review and notification phases of the conference process. The Conference Management System architecture is decomposed into four actors: Submission Manager, Review Manager, Notification Manager and Reviewer. Each of these actors is linked to the system through an is-part-of relationship (see Figure 4). The Submission Manager is responsible for handling the submission phase of the process. The Review Manager is responsible for distributing the set of submitted papers to at least n reviewers according to their research area. The Notification Manager is in charge of handling the notification phase process. The Reviewer actor is responsible for evaluating a paper proposal according to the reviewer preferences and skills.

The system architecture illustrated in Fig. 4. Presents a global view of software components. In order to realize such a virtual conferencing system, some automatic tools are required: some automatic tools are required:

1. The Presentation Synchronization and Recording Tool:

Conference presentations are conducted using slides and on-site presentations with questions and answers. These events can be recorded in video form. The synchronization tool allows a presenter to use a video camera and a presentation tool (such as PowerPoint), to record talks. Video presentations are synchronized with slide page changes.

2. The Conference Presentation Editing Tool and Database:

Conference presentations can be edited before they are stored in the Conference Presentation Database. Whether a presentation is annotated before it is stored in the database, the presentation is usually combined with questions and answers, which are also video clips recorded while the conference is being held.

Fig. 4. The software components of virtual conferencing system. T.K. Shih et al. Video proceedings of the virtual conference can be published based on the video records stored in the conference database.

3. The Mobile Storage/Server Replication Manager:

A conference can be held in real-time as well as retrieved from the conference presentation database. The mobile storage/server replication manager duplicates presentation slides, video presentations, or even question and answer video records to an appropriate location in a nomadic network environment. Replication reduces the network transmission load, thus, saves the bandwidth for video communications. The manager includes a mobile video server, which deals with realtime video transmission.

4. The Virtual Conferencing Administration System:

A virtual conference registration system allows participants to select presentation sessions. The registration system and the conference manager provide schedules and participant list of all the presentations. Therefore, the replication manager can arrange the communication bandwidth ahead of time

5. The MPEG-2 Video Communication Tool:

An MPEG-2 based video tool allows participants and the presenter to discuss topics in a real-time manner. Note that the virtual conferencing system supports both on-line virtual conferencing

and off-line conferencing. The on-line conference has session presentations transmitted without editing. However, the transmission records can be stored in the conference presentation database for publication of video proceedings. On the other hand, session presentations can be pre-arranged by presenters and conference managers. In this case, presentations can be annotated with simple drawings as well as video presentations. Whether a presentation is annotated, questions and answers can be recorded as video records, which can be included in the video proceedings *Wang (et.al) 2001*.

3.1.2 Data Flow Diagram

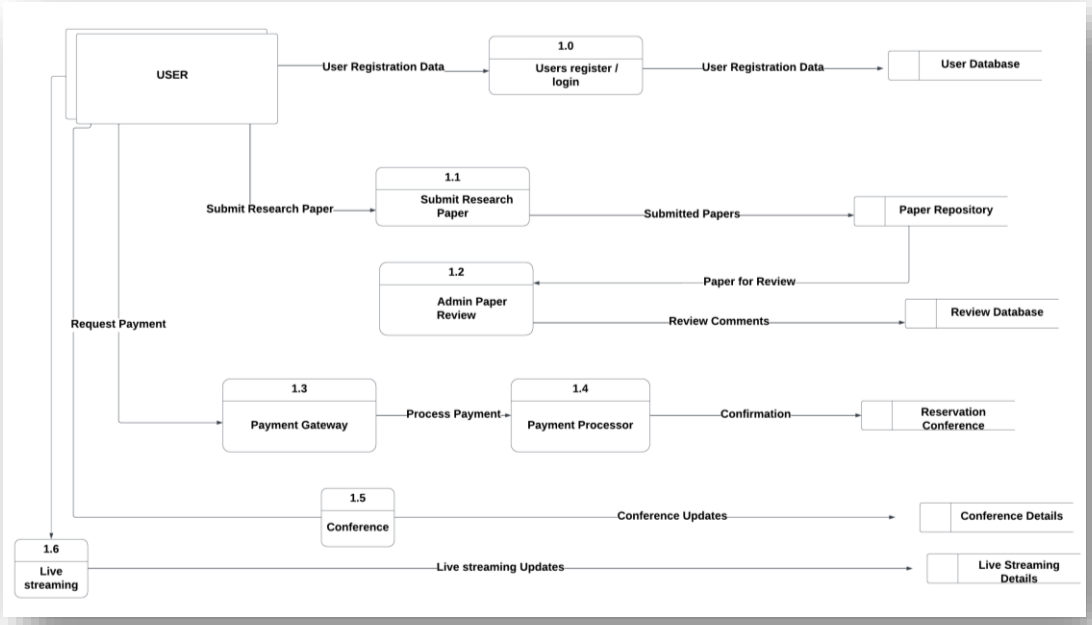

Figure 5: Data flow Diagram of Conference Portal System

In figure 5, It displays the data flow diagram as well as interacts users with the web-based system. Beginning with the login page, which identifies the user roles and directs them to the proper page suited to their participation in the conference. In addition, this data flow diagram illustrates the relationship between the system's operations and its many participants.

3.3.3 Software Development Life Cycle

The process of developing the proposed Research Conference Portal for the attendees and conference organizer of CITE Department in PHINMA-Saint Jude College Manila.

Agile methodology is a project management framework that breaks projects down into several dynamic phases, commonly known as sprints. The Agile framework is an iterative methodology. After every sprint, teams reflect and look back to see if there was anything that could be improved so they can adjust their strategy for the next sprint (Laoyan, 2022). Agile is an approach to project management that centers around incremental and iterative steps to completing projects. The incremental parts of a project are carried out in short-term development cycles. The approach prioritizes quick delivery, adapting to change, and collaboration rather than top-down management and following a set plan (Coursera, 2023). Agile approaches involve continuous feedback, giving team members the chance to adapt to problems as they appear and stakeholders the chance to communicate consistently. Although the Agile technique was initially developed for software development, it is now extensively used to carry out many various kinds of projects and manage organizations.

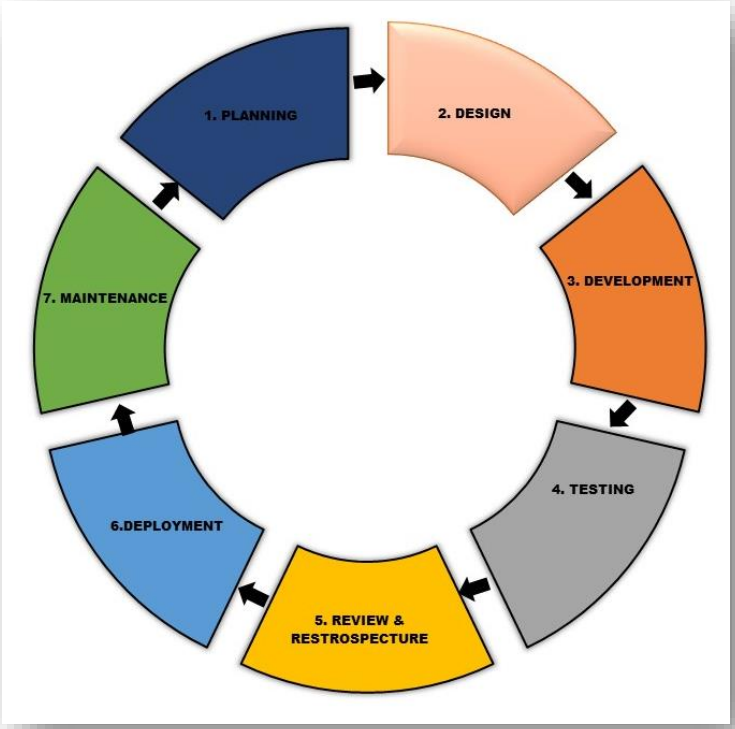

Figure 6: The development life cycle of the system

The proponents decided to use the Agile Development Life Cycle, since they want to be adaptive and accept improvement throughout the development of the system. They are willing to accept changes and iteration from the suggestions and feedback of users after testing it for the

betterment in developing the system. And to provide users with a user-friendly, understandable, time-saving and a better service throughout the development process of their event.

3.3.4 System Flowchart

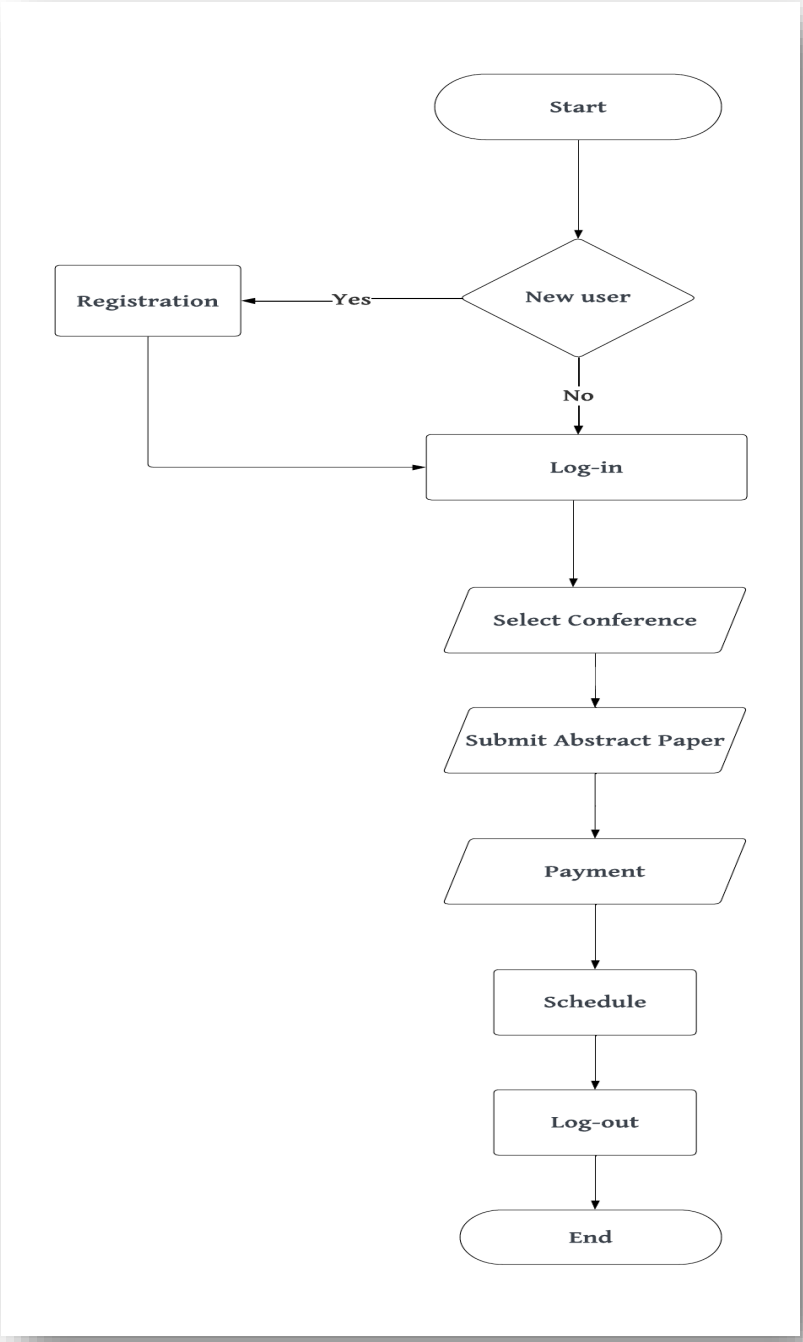

Figure 7: Student/Participant

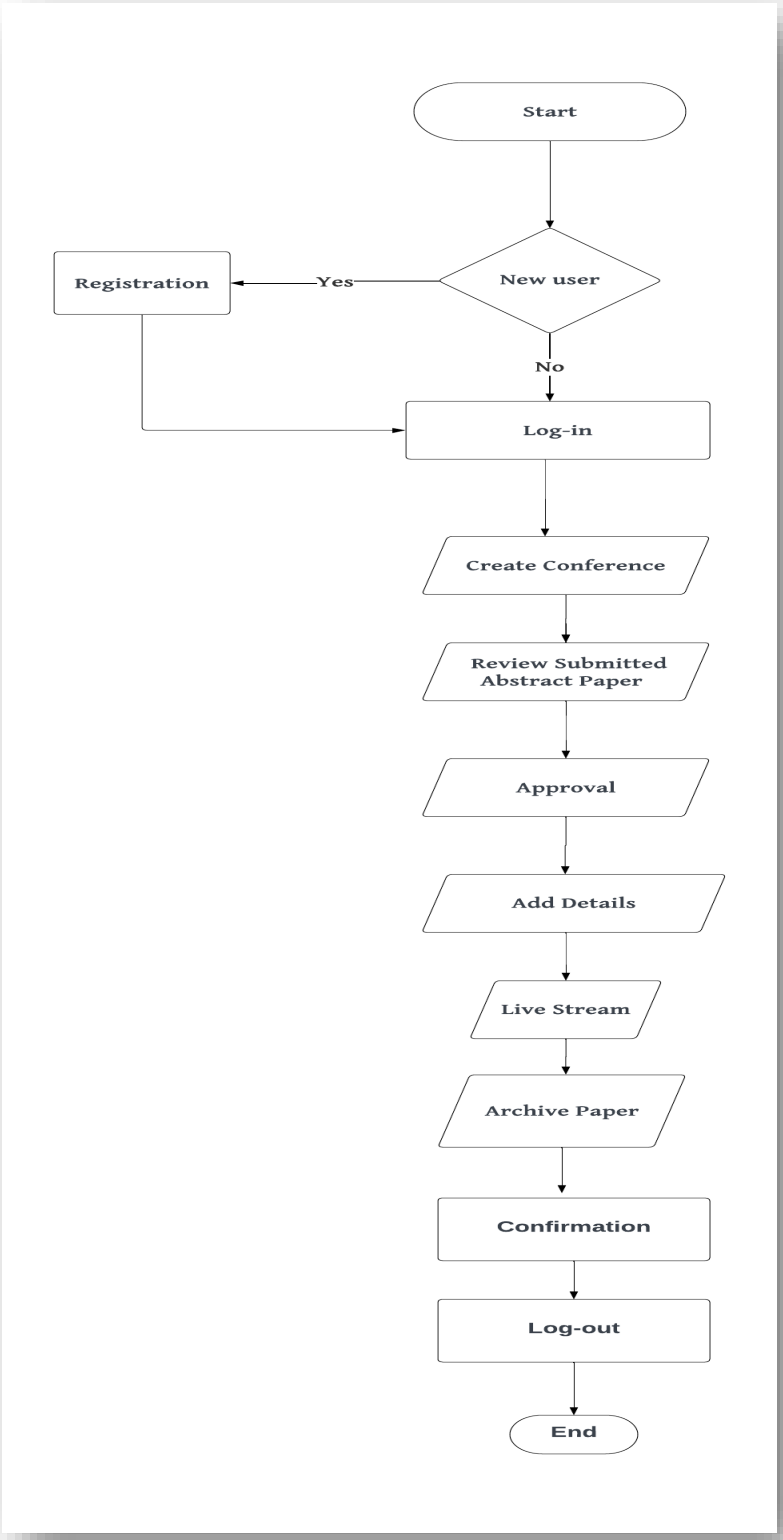

Figure 8: Admin

3.3.5 Entity Relationship Diagram

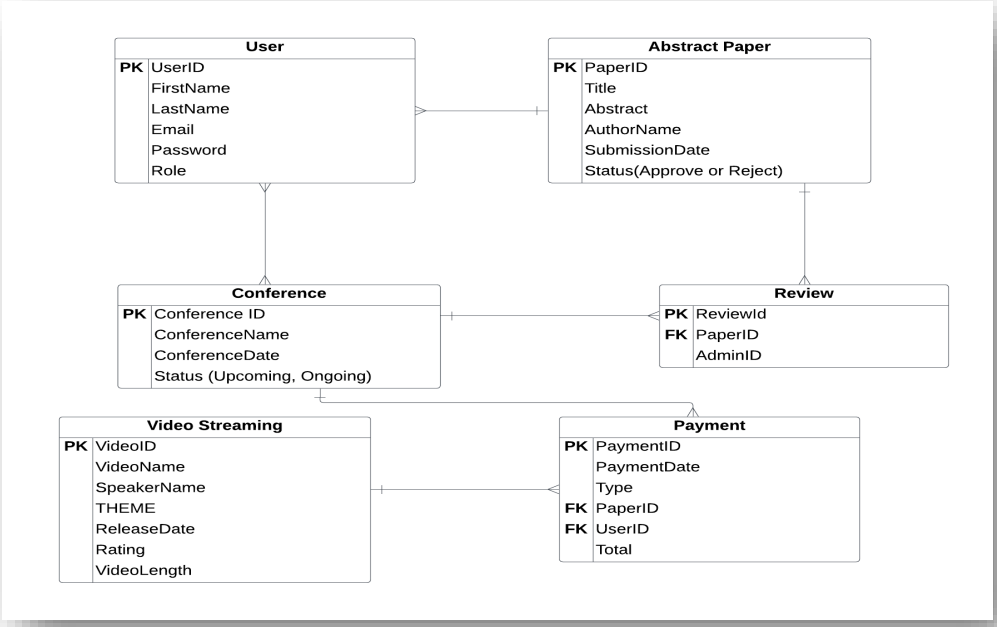

Figure 9: Entity Relationship Diagram of Conference Portal System

3.3.6 Use Case Diagram

Use Case Diagrams of the Proposed System

The Admin of the Research Conference Portal with Archiving monitors the website of the conference details. From initializing conference details to moderating submissions abstract paperr, managing financial transactions, and communicating essential notifications, the admin's role is multifaceted. Additionally, they are responsible for maintaining the integrity and security of the platform, ensuring that user data is protected and the portal remains resilient against potential cyber threats.

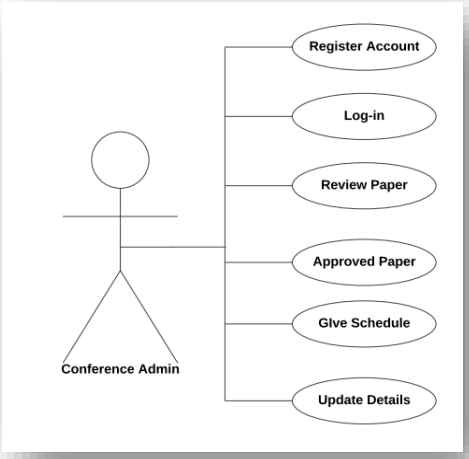

Diagram 1: Conference Administration

Student and participants of the Research Conference Portal with Archiving, they might aim to network, share their ideas research, engage in critical discourse, or simply update themselves on the latest developments in their field. Together, students and participants create a learning and expertise, ensuring that the conference portal becomes an interesting ideas, discussions, and collaborative growth. They attend conferences to stay updated on the latest developments in their field, connect with like-minded individuals, and contribute their own expertise.

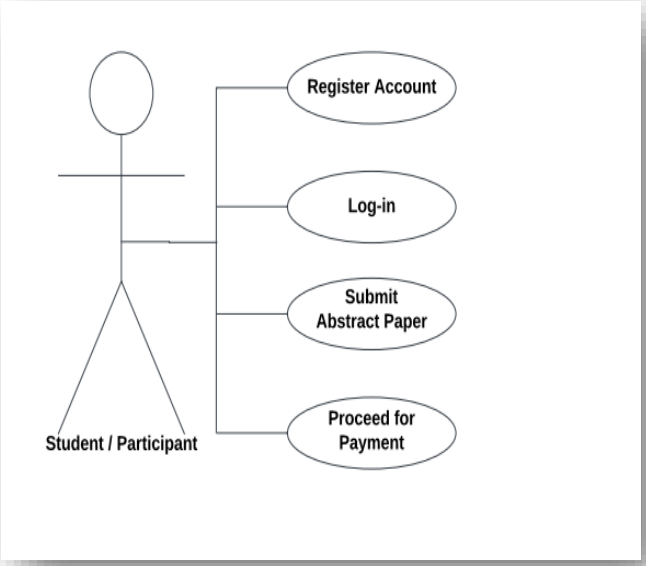

Diagram 2: Student and Participants

3.3.7 System Screenshots

Home page

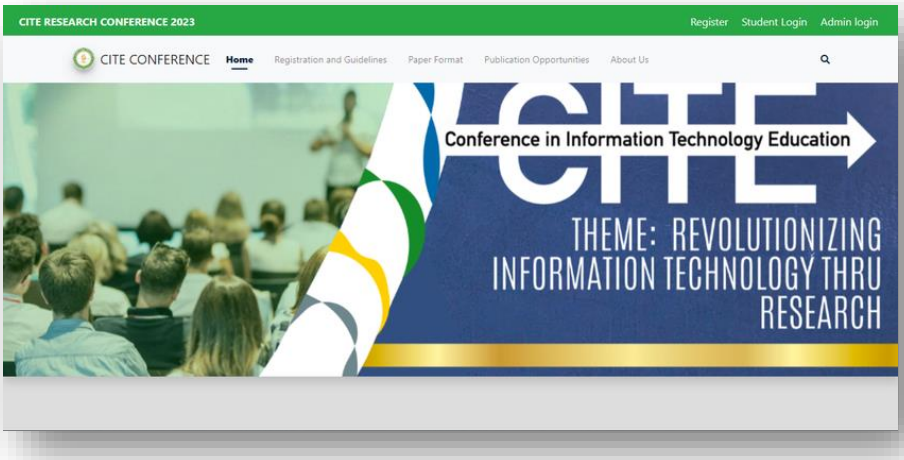

Figure 10: User Interface of Homepage

3.2 Statistical Treatment of Data

3.2.1 Data Gathering Technique

The researchers performed a survey in which they distributed questionnaires on a specific topic and had the respondents fill them out using Google forms. The study used a quantitative approach, a survey, because it was based on respondents. Questionnaires were given to selected responders who are IT students at PHINMA-Saint Jude College Manila.

Interview

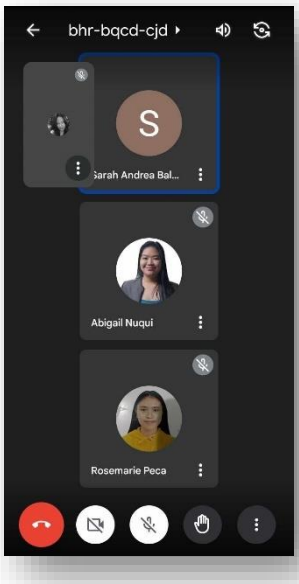

Figure 11: Interview with the IT Professor Ms. Sarah Viana via Google mee

Sample Interview Questionnaire:

Interview Questionnaire:

1. What are the key challenges you face in managing conferences, abstract submission and review process?

- submit only without review process

2. Are there any existing systems, tools or databases that the new conference management portal system needs to integrate with?

- no existing system

3. How do you currently plan conference schedules, sessions, and speakers?- it depends sa dami ng participants

4. How do authors currently submit their conference abstracts?

-Send a message to the conference admin/organizer through email

5. Can you explain the current process of attendee registration and payment handling?

- They currently used a google form for registration and through bank or cashier for payment

6. What issues, if any, have you faced regarding payment processing and attendee data management?

7. How do you communicate updates and announcements schedule changes or speaker update to conference participants and attendees?

8. What kind of technical infrastructure do you have in place for hosting a conference management portal system?

-None

9. Are there any specific security requirements or data protection measures that need to be considered?

SSL, Login, Paypal for payment

10. Is there any additional information or specific requirements you would like to share regarding the conference management portal system?

Sample Survey Questionnaire:

Research Conference Portal User Feedback Survey

Thank you for participating in this survey! Your feedback is valuable to us as we strive to improve our research conference portal. Please take a few minutes to answer the following questions.

1. General Information:
Name (Optional):
Year level:

2. Portal Usage:
- How did you learn about the research conference portal?
  - [ ] Email invitation
  - [ ] Social media
  - [ ] University/institution announcement
  - [ ] Other (please specify): _____

- How often did you use the portal during the conference?
  - [ ] Daily
  - [ ] Few times a week
  - [ ] Once a week
  - [ ] Rarely
  - [ ] Did not use

3. User Experience:
- Ease of Navigation:
  - [ ] Very Easy
  - [ ] Easy
  - [ ] Neutral
  - [ ] Difficult
  - [ ] Very Difficult

- Content Relevance:
  - [ ] Highly Relevant
  - [ ] Relevant
  - [ ] Neutral
  - [ ] Irrelevant
  - [ ] Completely Irrelevant

- Was the portal user-friendly?
  - [ ] Yes
  - [ ] No
  - [ ] Not sure

4. Features:

- Which features of the portal did you find most useful? (Select all that apply)
  - [ ] Event schedule and agenda
  - [ ] Virtual exhibition hall
  - [ ] Live streaming of sessions
  - [ ] Networking opportunities
  - [ ] Q&A and discussion forums
  - [ ] Downloadable resources (papers, presentations, etc.)
  - [ ] Other (please specify): _____

- Were there any additional features you wished the portal had?

5. Technical Issues:

- Did you encounter any technical issues while using the portal?
  - [ ] Yes
  - [ ] No
  - [ ] If yes, please describe the issue(s): _____

6. Overall Satisfaction:

- How satisfied are you with your overall experience using the research conference portal?
  - [ ] Very Satisfied
  - [ ] Satisfied
  - [ ] Neutral
  - [ ] Dissatisfied
  - [ ] Very Dissatisfied

- Would you recommend the portal to your peers and colleagues?
  - [ ] Yes
  - [ ] No
  - [ ] Maybe

7. Comments and Suggestions:
- Please share any additional comments, suggestions, or feedback about the research conference portal:

Thank you for taking the time to complete this survey. Your input is invaluable in helping us enhance the research conference portal for future events.

3.2.2 Setting of the Study

The study will take place in PHINMA Saint Jude College Manila which is located at Dimasalang cor, Don Quijote St, Sampaloc, Manila, 1008 Metro Manila. The scope of the system is only for the CITE department where the office is located at Dimasalang building second floor room 235. PHINMA SJC Manila has its own venues for research conferences inside the school which can be used during the execution of the conference. They have the multipurpose room (MPR) which is located at Silang Building second floor Room 2. It was a great place for this kind of event. It has a big white board/screen for the projection of presentation together with chairs and tables to be used by all participants, researchers, conference faculty, teachers, and guests.

This project is to also analyze the business environment in PHINMA- Saint Jude College in which the conference industry functions increasingly popular worldwide, conference management is integrated in educational plans. Nowadays there are various and common types of conference for hosting events. Conferences have an impact as a part of society development.

The system's infrastructure includes laptop, desktop, mobile phone, and network connection in accessing and using the system. The data of all users that was inputted into the system will be stored in one place which is the database. The proponents use the MySQL database for storing this data and for keeping all of the information safe inside the system.

Moreover, the proponents decided upon conducting the research in PHINMA Saint Jude College Manila to help the school in terms of a system about research conferences portal, which the school didn't have. The CITE department of the said school conducted a research conference last year 2023 in which they encountered many challenges along the event process. That's why the proponents decided to address these problems and decided to develop the system.

Figure 12: Illustrates the map of PHINMA- Saint Jude College Manila

3.2.3 Subject of the Study

In this survey study, IT student of each year in PHINMA-Saint Jude College Manila will be the selected randomly respondents in the process. The subjects and respondents of this study from first year to 4th year student of PHINMA- SJC enrolled in Manila. A Total of seven hundred eight students (708).

CITE STUDENTS	
SCHOOL YEAR	2023-2024
1st Year	408
2nd Year	209
3rd Year	51
4th Year	40
TOTAL	708

Table 1: Illustration of the Population of IT student in PHINMA- SJC Manila

As seen in table 1 a total of 708 student will be chosen randomly to participate in the survey.

3.2.4 Sampling Technique

According to McCombes (2023), there are two primary types of sampling methods that you can use in your research: probability and non-probability sampling methods. Probability sampling involves random selection, allowing you to make strong statistical inferences about the whole group. Probability sampling method has four main types such as simple random sampling, systematic sampling, stratified and cluster sampling. While non-probability sampling involves a collection of feedback based on a researcher or statistician's sample selection capabilities and not on a fixed selection process (Fleetwood, 2023). Non-probability sampling method has four main types such as convenience sampling, purposive sampling, snowball sampling and quota sampling.

The proponents decided to choose the simple random sampling to be used during the collection of data from the respondents. This is to easily collect information from the respondents without calculating any numbers of respondents which the proponents get from the students of the

CITE department. A simple random sampling is a randomly selected subset of the population (Thomas, 2020). A subset of a statistical population called a simple random sample is one in which each member has an equal chance of being chosen. A straightforward random sample is intended to offer an objective depiction of a group (Hayes, 2023). Of all the probability sampling techniques, this one is the easiest to understand because it only needs one random selection and little prior population knowledge. Any research conducted with this sample should have high internal and external validity and be less likely to be biased by factors like sampling bias and selection bias because randomization is used.

The proponents will get a random number of students in each year level and that will be the total respondents of the system. The population will be all the students of CITE department of PHINMA SJC Manila, which has a total population of 708 students enrolled for the school year 2023-2024. The sample size of population to be selected will be 50 students from all grade levels that will be randomly selected for data collection.

3.2.5 Statistical Treatment of Data

The respondents' answer to the questionnaire by the random IT students in PHINMA- SJC Manila will be analyzed with the data requirements of the research study. The survey with a 5 (or 7) point Likert Scale which is used by respondents to rate the degree to which ranging from agree to strongly disagree with a statement and allowing the respondents to select answers ranging across of choices to gain more insight and opinion. Likert Scale frequently used in education research Sulivan et. al (2013). After the data has been gathered, the response with the most votes will be considered the respondents' answer.

For various study structures, several statistical treatments are considered. The following are relevant in this context:

Sample size determination

The technique of electing the number of observations to include in a sample. The sample size is an important feature of any study or investigation in which the aim is to make inferences about the population from a sample. In general, the sample size used in a study is determined based on the cost of data collection, and based on sufficient statistical power. In advanced studies there may be

several different sample sizes involved in the study: for example, in a survey sampling if population is heterogeneous involving stratified sampling there would be different sample sizes for each population. Singh et. al (2014).

Sampling Method

The selection of a subset of individuals from within a population to estimate the characteristics of whole population. The two main advantages of sampling are the faster data collection and lower cost. (Kish 1965, Robert 2004). Each observation measures one or more properties of observable subjects distinguished as independent individuals. In business research, medical research, agriculture research, sampling is widely used for gathering information about a population Singh et. al (2014).

Statement

Academic detailing is a useful form of education that aligns providers' prescribing behavior with evidence-based practice.

Strongly Disagree	Disagree	Neutral	Agree	Strongly Agree
1	2	3	4	5

Table 2: Illustration of the Likert Scale

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